Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/11/2023
PI Name:	Hahn, Inseob Ph.D.		
Project Title:	Investigation of equilibration near the liquid-gas critical point in microgravity utilizing DECLIC/ALI facility		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	inseob.hahn@jpl.nasa.gov	**Fax:**	FY 818-393-4878
PI Organization Type:	NASA CENTER	**Phone:**	818-354-7999
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	4800 Oak Grove Dr.		
PI Address 2:	79-24		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	12/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
Contact Email:	john.l.callas@jpl.nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2024, per PI (Ed., 10/14/23) NOTE: End date changed to 9/30/2023, per J. Callas/JPL (Ed., 11/14/2022) NOTE: Changed JPL Task Plan number to 71-15884 per U. Israelsson/JPL (Ed., 5/27/2020) NOTE: End date changed to 9/30/2022, per PI (Ed., 2/27/2020) NOTE: End date changed to 9/30/2019, per PI (Ed., 11/17/17); changed to 10/1/2019 to accommodate final reporting (Ed., 10/4/18) NOTE: End date changed to 10/01/2017 (Ed., 11/14/16) NOTE: Extended for reflight to 9/30/2017 (per PI and U. Israelsson/JPL)-Ed., 5/12/14 NOTE: Extended to 3/31/2014 per PI (Ed., 10/21/13)		
Key Personnel Changes/Previous PI:	Per the Principal Investigator: Fang Zhong, Ph.D is not currently with the project and has been removed from the list of Co-Investigators (Ed., 10/14/23).		

COI Name (Institution):	Beysens, Daniel Ph.D. (Commissariat a l'Energie Atomique at Grenoble and Ecole Supérieure de Physique et Chimie Industrielle) Garrabos, Yves Ph.D. (Centre National de la Recherche Scientifique (CNRS)-)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory) Lecoutre, Carole M.S. (Centre National de la Recherche Scientifique (CNRS)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory)
Grant/Contract No.:	JPL Task Plan number 71-15884 ; JPL Task Plan number 71-15890
Performance Goal No.:	
Performance Goal Text:	
Task Description:	A key feature of a continuous phase transition in many physical systems in Nature is scale-invariance in fluctuations of the order parameter, which implies continuity between large and small scale fluctuations associated with the transition. Scale-invariance occurs in many systems that are still not completely understood, such as the universe at the onset of the big bang as currently envisioned by cosmologists, the clustering of galaxies, massless fields in elementary particle physics, the distribution of earthquakes, and turbulence in fluids and plasmas. In each of these examples there is a wide range of scales over which some phenomena vary as a power-law of the scale. The Nobel Prize winning Renormalization Group (RG) theory is the most successful approach thus far developed to derive and explore the relations between the characteristics of a system as viewed on different length scales. Many scientific investigations using simple fluid systems have been performed on the ground. The predictions of the RG theory have been reasonably well validated. Nevertheless, the theory is based on "seemingly innocuous" assumptions that will stand or fall based on more sophisticated experiments. Experimental studies of critical phenomena using simple fluids on Earth have been very difficult to perform due to the gravity induced density gradient. The DECLIC (Dispositif pour l'Etude de la Croissance et des LIquides Critiques)-EVO (Evolutionary) facility is an ideal platform to study the critical phenomena without gravity effects. Latest operations of the legacy DECLIC Alice-Like-Insert (ALI) have shown that the turbidity measurements could be performed very close to the critical point, taking advantages of the high-quality thermal and optical environment of the ALI insert and the microgravity condition of the International Space Station (ISS). The ALI cell was originally filled with sulfurhexafluoride (SF6) sample at slightly off-critical density for studying the boiling phenomena. The turbidity measurement at the off-critical density (~1.0% off from the critical density) was used to test and improve an Ising-like classical-to-critical crossover model. We proposed a re-flight of the Alice-Like-Insert (ALI-R) using a new cell with the same SF6 sample at much closer to the critical density and with an optimal geometry. The ALI-R will measure the effects of the non-zero value of the Green-Fisher exponent of the liquid-gas critical point. A measurement of the turbidity of the SF6 sample with a precision of 0.1% at 1 mK above the critical temperature (Tc=318.733 K) could test the theoretical predictions of Green-Fisher exponent with a unprecedented precision.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide a unique opportunity to measure experimental properties of a fluid near its liquid-gas critical point without gravitational effects and to refine the theoretical equation of state models. The critical and supercritical state of fluids exhibit unique properties. For instance, the fluid near the critical point and supercritical state shows very large compressibility. The use of these fluids, for instance, for the storage of cryogenic propellant, under normal gravity and/or reduced gravity conditions raises fundamental questions regarding fluid dynamics, heat transfer, interfacial phenomenon. These experimental works will enhance the understanding of basic physics regarding the fluids near critical state.
Task Progress:	The Alice-Like Insert (ALI-R) was successfully integrated with the Dispositif pour l'Etude de la Croissance et des Liquide Critiques (DECLIC-EVO) aboard the International Space Station (ISS). And the Principal Investigator (PI) team successfully completed three ALI-R science sequences. The main science sequence was to perform the light transmission measurement (turbidity) of the sample fluid (SF6) near its liquid-gas critical point between 0.1 mK < T_{c}-T < 3 K. We have analyzed the data to determine the turbidity of the fluid. We also completed the development of a new universal, parametric equation-of-state model, named the Crossover Master Model (CMM). A manuscript is submitted to a peer-review journal. A DECLIC workshop 2022 was held at the French Space Agency (Centre national d'études spatiales/CNES), Toulouse, France. The PI team presented the ALI-R progress at the workshop. The CNES engineering team successfully modified the main flight computer boot sequence to bypass an electronics board anomaly. The in-orbit troubleshooting runs were performed successfully in June 2023. And two additional ALI-R science operations are planned (starting in October, 2023).
Bibliography Type:	Description: (Last Updated: 01/02/2025)
Articles in Peer-reviewed Journals	Oprisan A, Morgado D, Dorf D, Zoppelt S, Oprisan SA, Hahn I, Garrabos Y, Lecoutre-Chabot C, Beysens D. "Transport properties of critical sulfur hexafluoride from multiscale analysis of density fluctuations." Front. Space Technol. 2022 Jun 20;3:883899. https://doi.org/10.3389/frspt.2022.883899 , Jun-2022